

Question number	Answer	Additional guidance	Mark
4(a)(i)	An answer that includes the following points: - an increase in the number of (DNA) {molecules / double helices} (1) - each (new molecule) consists of one {parent / original / old} strand and one new strand (1)	ACCEPT {two / new} molecules are made (from one molecule)	(2)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	(DNA) helicase / polymerase / ligase (1)	ACCEPT swivelase / untwistase / topoisomerase	(1)

Question number	Answer	Additional guidance	Mark
4(a)(iii)	An explanation that includes the following points: because it results in genetically identical (daughter) cells (1)	ACCEPT same {genetic information /	

	that will have the same {structure / function} (as the parent cell) (1)	genotype / alleles / DNA sequence}	(2)
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4(b)	<table> <tr> <th rowspan="2">Stage</th><th colspan="3">Percentage of the DNA molecules containing</th></tr> <tr> <th>heavy nitrogen only</th><th>light nitrogen only</th><th>both heavy and light nitrogen</th></tr> <tr> <td>1</td><td>100</td><td>0</td><td>0</td></tr> <tr> <td>2</td><td>0</td><td>0</td><td>100</td></tr> <tr> <td>3</td><td>0</td><td>50</td><td>50</td></tr> </table>	Stage	Percentage of the DNA molecules containing			heavy nitrogen only	light nitrogen only	both heavy and light nitrogen	1	100	0	0	2	0	0	100	3	0	50	50	N.B. mark the rows not the columns DO NOT ACCEPT empty boxes	(3)
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2	0	0	100																			
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